3 Which person in the following situations is experiencing weightlessness?

A An athlete clearing the bar in a high jump.
B A swimmer floating in water.
C A parachutist falling at terminal speed.
D A pilot in a hovering helicopter.

Ans: A

To experience weightlessness, the person should experience only gravitational force acting on him.

Option B: the person also experiences a buoyant force from the water.
Option C: the person also experiences an upward force from the parachute harness.
Option D: the pilot experiences an upward normal reaction from the seat he is sitting in